

Digital Forensics Laboratory

EXPERIMENT NO: 05

TITLE: Digital Forensics Investigation Using Autopsy: Case Creation and Evidence Import

1. AIM

To use the Autopsy digital forensics platform to create a new forensic case, import digital evidence (disk image files), configure ingest modules, and perform basic forensic analysis and reporting.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows, Linux, or macOS
- **Forensics Tool:** Autopsy Digital Forensics Platform
- **Evidence Files:** Disk image files (4Dell Latitude CPi.E01, 4Dell Latitude CPi.E02) downloaded from the provided forensic repository.

3. THEORY

Autopsy is a premier open-source digital forensics platform and graphical interface to The Sleuth Kit® and other digital forensics tools. It is widely used by law enforcement, military, and corporate examiners to analyze and extract critical data from digital devices.

In a standard forensic investigation, raw data or disk images (such as .E01, .dd, or .raw formats) are imported into a securely managed "Case". Autopsy utilizes various **Ingest Modules** (like File Type Identification, Keyword Searching, Timeline Analysis, and Hash Lookups) to process this evidence in the background, automatically categorizing artifacts such as web history, file system metadata, and communication records for the examiner to review.

4. PROCEDURE

Step 1: Installation & Preparation

1. Download and install the Autopsy application based on the host operating system.
2. Download the required evidence files (4Dell Latitude CPi.E01) to a secure local directory.
3. Launch the Autopsy application to view the welcome screen.

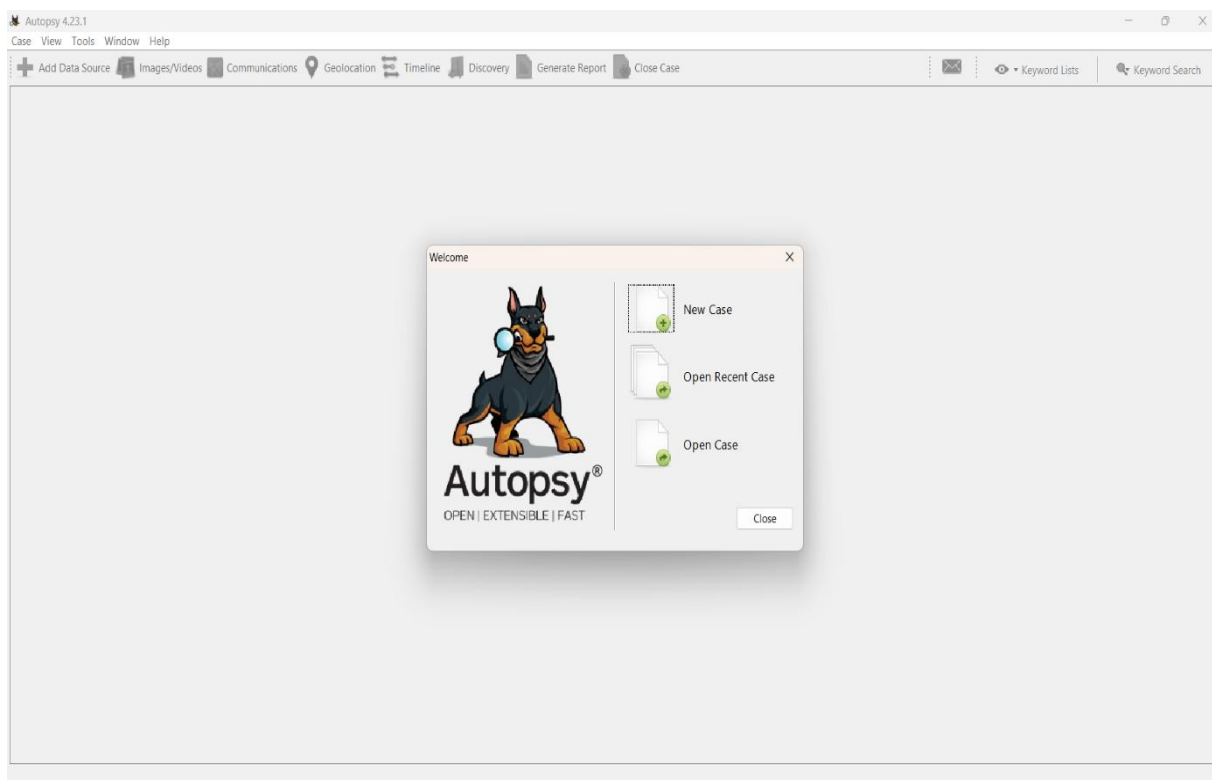


Figure 1: Autopsy startup menu and tool interface

Step 2: Starting a New Case

1. Click on **New Case** from the startup menu.
2. Enter the case name (e.g Surya-Investgated) and specify the base directory where the case data will be stored.
3. Click **Next** to proceed to the optional information.

- Fill in the required case details, including the case number and examiner's name (Surya Prakash T), then click **Finish**.

The screenshot shows the 'New Case Information' dialog box with the 'Case Information' tab selected. The 'Steps' list on the left shows '1. Case Information' and '2. Optional Information'. The 'Case Information' section contains the following fields:

- Case Name:
- Base Directory:
- Case Type: ☒ Single-User ☐ Multi-User
- Case data will be stored in the following directory:

At the bottom, there are buttons: < Back, Next >, Finish, Cancel, and Help.

Figure 2: Creating a new forensic case and setting the base directory

The screenshot shows the 'New Case Information' dialog box with the 'Optional Information' tab selected. The 'Steps' list on the left shows '1. Case Information' and '2. Optional Information'. The 'Optional Information' section contains the following fields:

- Case Number:
- Examiner Name:
- Examiner Phone:
- Examiner Email:
- Examiner Notes:
- Organization analysis is being done for:

At the bottom, there are buttons: < Back, Next >, Finish, Cancel, and Help.

Figure 3: Entering Optional Information including Examiner details

Step 3: Adding a Data Source

1. When prompted to add a data source, choose the type of data source (e.g., Disk Image or VM file).
2. Browse to the local directory and select the downloaded evidence file (4Dell Latitude CPi.E01).
3. **Configure Ingest Modules:** Select the specific analysis modules required for the investigation.
4. Click **Next** to start the analysis.

The screenshot shows the 'Add Data Source' window. On the left, a 'Steps' list indicates the current step is '3. Select Data Source'. The main panel, titled 'Select Data Source', contains the following fields and options:

- Path:** A text box containing 'C:\Users\surya\Downloads\4Dell Latitude CPi.E01' with a 'Browse' button to its right.
- Ignore orphan files in FAT file systems:** An unchecked checkbox.
- Time zone:** A dropdown menu showing '(GMT+ 5:30) Asia/Calcutta'.
- Sector size:** A dropdown menu showing 'Auto Detect'.
- Bitlocker Password (optional):** An empty text box.
- Hash Values (optional):** Three empty text boxes for MD5, SHA-1, and SHA-256.

A note at the bottom states: 'NOTE: These values will not be validated when the data source is added.' At the bottom right, there are five buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

Figure 4: Importing the .E01 evidence file data source

Step 4: Initial Analysis and Overview

1. Monitor the **Ingest Progress** in the lower-left corner of the interface as Autopsy processes the data source.
2. Once processing yields results, use the **Tree Viewer** on the left pane to explore different categorized aspects like File System, Deleted Files, Data Artifacts, etc.

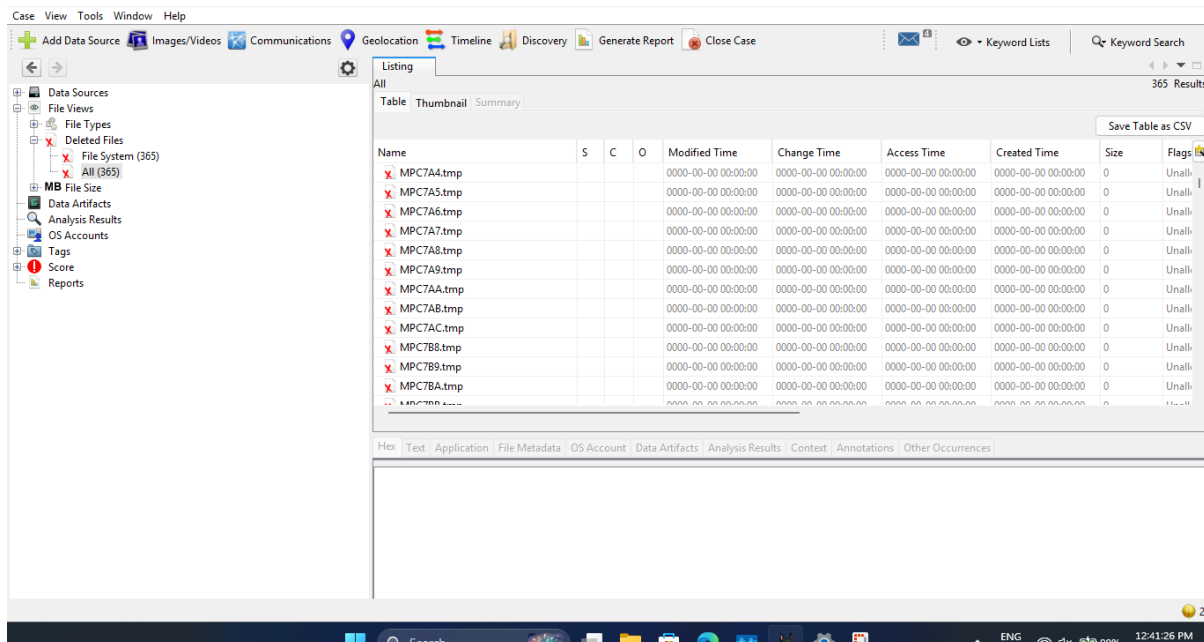


Figure 5: Exploring categorized forensic artifacts and deleted files in the Tree Viewer

Step 5: Detailed Analysis

1. **File Analysis:** Navigate through files and folders under the File Types or File System section. Open, view, or extract files for further examination.
2. Select an individual file (e.g., audio files under the Audio category) to view its hex, text, application, or file metadata in the lower pane.

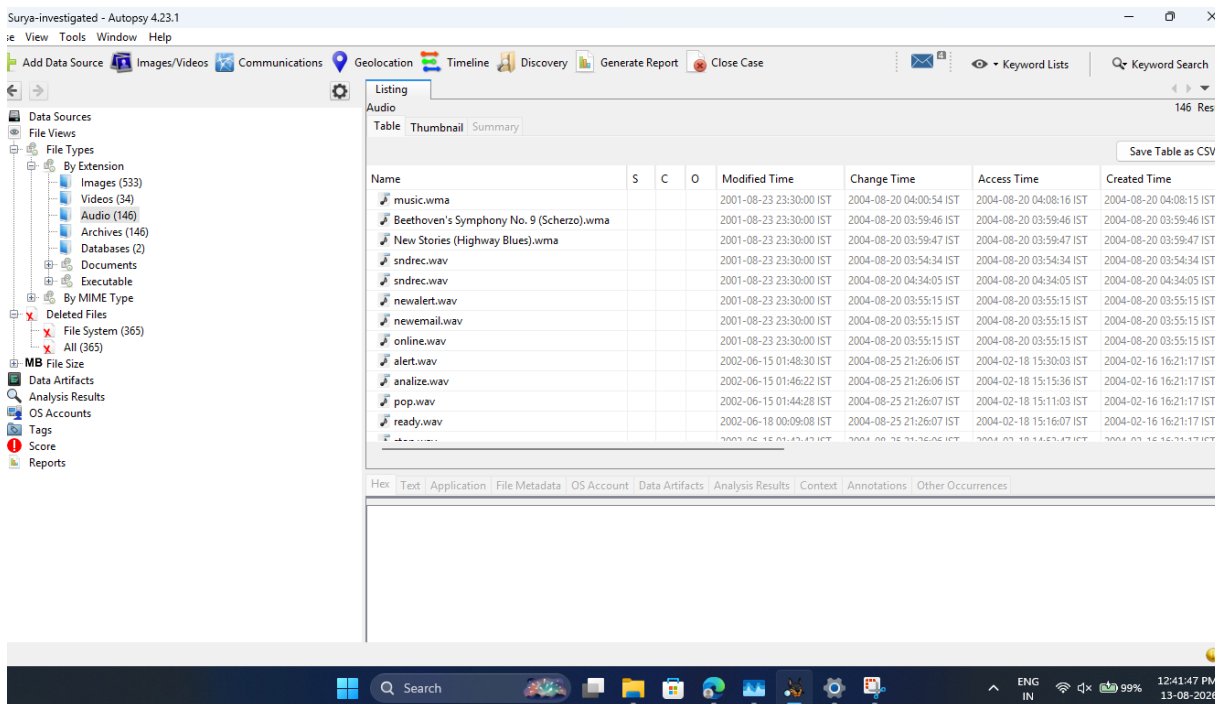


Figure 6: Detailed analysis and viewing extracted files (e.g., Audio playback and metadata)

Step 6: Reporting and Case Closure

1. Click on **Generate Report** from the main toolbar.
2. Choose the desired report format (e.g., Excel Report, HTML Report) under the Select and Configure Report Modules window.
3. Click **Next** and select the specific result types to include (e.g., Accounts, E-Mail Messages, etc.).
4. Review the generated report to ensure all relevant information and exported artifacts are accurately documented.
5. Once the investigation is complete, close the case within Autopsy and archive all data according to organizational policies.

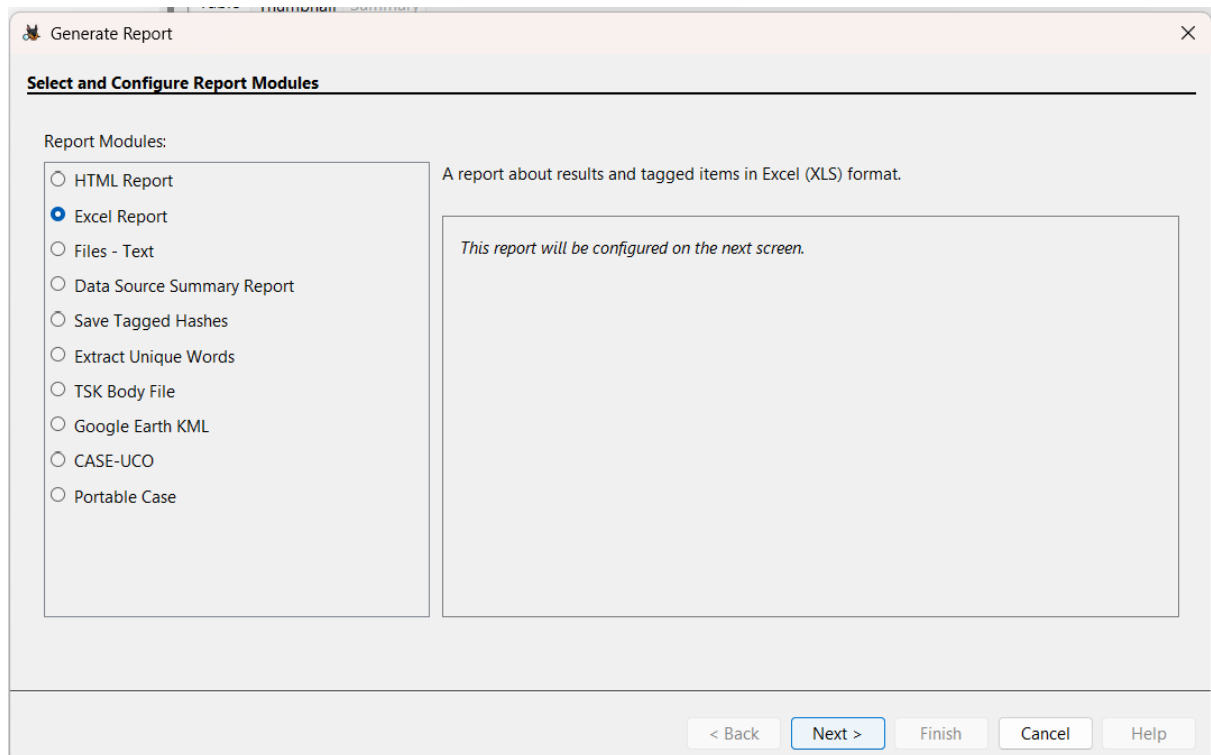


Figure 7: Selecting the report module format

	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
30:00 IST	2004-08-20 04:00:54 IST	2004-08-20 04:08:16 IST	2004-08-20 04:08:15 IST	3492199	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Documents and Settings/All Users/Documents/My Music/music.wma
30:00 IST	2004-08-20 03:59:46 IST	2004-08-20 03:59:46 IST	2004-08-20 03:59:46 IST	613638	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Documents and Settings/All Users/Documents/My Music/Sample Music/Beethoven's Symphony No. 9 (Scherz
30:00 IST	2004-08-20 03:59:47 IST	2004-08-20 03:59:47 IST	2004-08-20 03:59:47 IST	760748	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Documents and Settings/All Users/Documents/My Music/Sample Music/New Stories (Highway Blues).wma
30:00 IST	2004-08-20 03:54:34 IST	2004-08-20 03:54:34 IST	2004-08-20 03:54:34 IST	58	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Documents and Settings/Default User/Templates/sndrec.wav
30:00 IST	2004-08-20 04:34:05 IST	2004-08-20 04:34:05 IST	2004-08-20 04:34:05 IST	58	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Documents and Settings/Mr. Evil/Templates/sndrec.wav
30:00 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	9306	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/MSN/MSNCoreFiles/newalert.wav
30:00 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	18052	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/MSN/MSNCoreFiles/newemail.wav
30:00 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	2004-08-20 03:55:15 IST	9306	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/MSN/MSNCoreFiles/online.wav
48:30 IST	2004-08-25 21:26:06 IST	2004-02-18 15:30:03 IST	2004-02-16 16:21:17 IST	11370	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Look@LAN/sounds/alert.wav
46:22 IST	2004-08-25 21:26:06 IST	2004-02-18 15:15:36 IST	2004-02-16 16:21:17 IST	75812	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Look@LAN/sounds/analyze.wav
44:28 IST	2004-08-25 21:26:07 IST	2004-02-18 15:11:03 IST	2004-02-16 16:21:17 IST	4486	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Look@LAN/sounds/pop.wav
09:08 IST	2004-08-25 21:26:07 IST	2004-02-18 15:16:07 IST	2004-02-16 16:21:17 IST	15920	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Look@LAN/sounds/ready.wav
43:42 IST	2004-08-25 21:26:06 IST	2004-02-18 14:53:47 IST	2004-02-16 16:21:17 IST	19040	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Look@LAN/sounds/stop.wav
30:00 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	9306	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Messenger/newalert.wav
30:00 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	18052	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Messenger/newemail.wav
30:00 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	9306	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Messenger/online.wav
40:32 IST	2004-08-20 03:55:25 IST	2004-08-20 03:55:24 IST	2004-08-20 03:55:24 IST	4454	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Messenger/type.wav
30:00 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	3986	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND18.WAV
30:00 IST	2004-08-20 03:54:59 IST	2004-08-20 03:54:59 IST	2004-08-20 03:54:59 IST	13024	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND38.WAV
30:00 IST	2004-08-20 03:54:54 IST	2004-08-20 03:54:54 IST	2004-08-20 03:54:54 IST	108607	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/PINBALL.MID
30:00 IST	2004-08-20 03:54:54 IST	2004-08-20 03:54:54 IST	2004-08-20 03:54:54 IST	28888	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/PINBALL2.MID
30:00 IST	2004-08-20 03:54:54 IST	2004-08-20 03:54:54 IST	2004-08-20 03:54:54 IST	55490	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND1.WAV
30:00 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	1226	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND104.WAV
30:00 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	1968	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND105.WAV
30:00 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	7754	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND108.WAV
30:00 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	890	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND111.WAV
30:00 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	824	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND112.WAV
30:00 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	4296	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND12.WAV
30:00 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	2004-08-20 03:54:55 IST	8034	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND13.WAV
30:00 IST	2004-08-20 03:54:56 IST	2004-08-20 03:54:56 IST	2004-08-20 03:54:56 IST	1290	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND131.WAV
30:00 IST	2004-08-20 03:54:56 IST	2004-08-20 03:54:56 IST	2004-08-20 03:54:56 IST	19282	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND136.WAV
30:00 IST	2004-08-20 03:54:56 IST	2004-08-20 03:54:56 IST	2004-08-20 03:54:56 IST	3002	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND14.WAV
30:00 IST	2004-08-20 03:54:56 IST	2004-08-20 03:54:56 IST	2004-08-20 03:54:56 IST	1046	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND16.WAV
30:00 IST	2004-08-20 03:54:56 IST	2004-08-20 03:54:56 IST	2004-08-20 03:54:56 IST	2090	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND17.WAV
00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND17.WAV
00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND181.WAV
30:00 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	5230	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND19.WAV
30:00 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	8650	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND20.WAV
30:00 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	9194	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND21.WAV
30:00 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	7376	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND22.WAV
30:00 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	12106	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND24.WAV
30:00 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	14600	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND240.WAV
30:00 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	2004-08-20 03:54:57 IST	20712	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND243.WAV
30:00 IST	2004-08-20 03:54:58 IST	2004-08-20 03:54:58 IST	2004-08-20 03:54:58 IST	25704	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND25.WAV
30:00 IST	2004-08-20 03:54:58 IST	2004-08-20 03:54:58 IST	2004-08-20 03:54:58 IST	7306	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND26.WAV
30:00 IST	2004-08-20 03:54:58 IST	2004-08-20 03:54:58 IST	2004-08-20 03:54:58 IST	20242	Allocated	Allocated	unknown	/img_4Dell Latitude CPLE01/vol_vo2/Program Files/Windows NT/Pinball/SOUND27.WAV

Figure 10:Filtered Audio Results for Export

5. RESULT

The digital forensics experiment was successfully completed. A new case was established in Autopsy, and the provided .E01 disk image evidence was successfully imported, analyzed using ingest modules, and documented via the reporting feature.